SAVEETHA
INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE CODE: CSA1407 **COURSE NAME: Compiler Design**

COURSE OUTCOME COVERED

CO2: Construct top-down and bottom up parsers using CFG for parsing conflicts and error recovery for efficient syntax tree generation. (BL3)

Assessment Tool Weightage: 10%

QUESTIONS:

Total Marks:50

Annexure A

SCENARIO BASED TASK

Code of Conduct:

I Ramya S S(192421226) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
Ramya S S

RUBRICS FOR CALCULATION:

Scenario based assessment

Criteria	Weight age (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and

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COURSE: CSA1407 COMPILER DESIGN

CO2 - AT3 - SCENARIO BASED TASK

① Scenario: Parsing table conflict in LL(1).

Ans: The LL(1) parser constructs its parse table using the FIRST and FOLLOW sets of the grammar. The FIRST set contains the terminals that can appear at the beginning of a production, while the FOLLOW set contains the terminal that can appear immediately after a non-terminal. These set help the parser decide which production rule to apply.

Decision:

The grammar should be modified by eliminating left recursion, applying left factoring and ensuring that the FIRST and FOLLOW sets do not overlap. This produces a conflict free LL(1) grammar suitable for deterministic parsing.

② Scenario: Incorrect Parse Tree Generation

The syntax analyzer constructs a parse tree by applying grammar rules to the input tokens. For the expression id+id-id, the parse tree determines the order in which operations are performed. If the grammar does not define operator associativity, the expression may be grouped incorrectly, leading to wrong evaluation results.

Decision:

The grammar should enforce left associativity for the '+' and '-' operators. As a result, multiple parse trees can be generated for the same expression, creating ambiguity.

$$E \rightarrow E+T \,/\, E-T \,/\, T$$

$$T \rightarrow id.$$

③ Scenario: Invalid Expression Handling

In the expression a+*b, the lexical analyzer correctly identify all tokens as valid identifiers and operators. However the syntax analyzer checks whether these tokens follow the grammar rules. After the '+' operator, the grammar expects an operand, but another operator '*' appears instead. Therefore, the parser cannot generate a valid parse tree.

Decision:

The parser should implement error recovery techniques such as panic-mode recovery, phrase level recovery, or synchronization using FOLLOW sets. These methods helps continue parsing and provide meaningful error messages.

resu Scenario : Ambiguity in conditional statements.

The grammar

$S \rightarrow$ if E then S

$S \rightarrow$ if E then S else S

$S \rightarrow a$

It is ambiguous because the parser cannot determine which if statement an else belongs to. This problem is called dangling else problem.

Decision:

The grammar should be rewritten using matched and unmatched statements.

$S \rightarrow M / V$

$M \rightarrow$ if E then M else M/a

$U \rightarrow$ if E then S | if E then M else U.

(5) Scenario : Operator precedence conflict

The grammar

$E \rightarrow E + E$

$E \rightarrow E * E$

$E \rightarrow id$

is ambiguous because it doesnot define operator precedence or associativity. For the expression id + id * id the parser may generate either (id + id) * id or id + (id * id).

Decision:

The grammar should be rewritten as

$E \to E + T / T$

$T \to T * F / F$

$F \to id$

Here:

* Multiplication has higher precedence than addition.
* Addition and multiplication are left associative.